Voting for a consensus with veto rights

Executive summary of the state-of-the-art note, version 0.1

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The question

A group must choose one option. **Anyone may propose as many options as they wish**, **every member judges all of them**, and **every member has a right of veto**. The procedure must reach the **best compromise** and must **never get the group stuck**. What does research say, and which procedure should be used?

Bottom line

1. **Do not give every member a hard veto.** Unanimity breeds delay, status-quo bias and holdouts; with the status quo as default it deadlocks in 86% of our simulated decisions with 11 members and 99.6% with 101.
2. **Use "Condorcet with a blocking quota" (QVC) with a quota of one third.** One ballot; an option rejected by at least a third of the members is out; the majority decides among the rest.
3. **Accept, and contain, its one price:** a blocking third can block strategically. Named, reasoned vetoes and at most one revision round keep that in check.

What the research says

1. **There is no perfect rule.** Every voting rule can be manipulated (Gibbard–Satterthwaite), and respecting the majority sometimes means overriding a large minority: designing a procedure means choosing a trade-off, and it should be explicit.
2. **Unanimity is rarely optimal.** Theory and experiments agree that it raises delay and the power of holdouts; majority voting with a *separate* veto, used only against unacceptable outcomes, does better (Bouton et al., 2018; Tsebelis, 2002).
3. **Vetoes should be held by coalitions, not individuals,** as in the Council of the European Union's blocking minority; consensus bodies such as the IETF treat an objection as an argument to address, not as a block.
4. **Open proposals require a "clone-proof" rule.** If anyone can add variants of an option, many rules reward flooding the agenda: three variants change the outcome of Borda's rule in up to 24% of our elections, of fallback bargaining in 36% and of the proportional veto core in 54%. Schulze-type Condorcet methods are immune.
5. **Condorcet methods give the best results when preferences have structure,** and a Condorcet winner (an option preferred by a majority to every other) exists in at least 96% of our simulated electorates, even with twelve options.

Recommended procedure: QVC(θ), with $\theta = 1/3$

1. **Nominate.** Anyone proposes any number of options. The status quo is on the ballot as an ordinary option, never as the default.
2. **Vote once.** Each member ranks all options (ties allowed) and marks those they find unacceptable; each veto is named and states a reason.
3. **Filter.** Options vetoed by at least a third of the members are excluded. If that excludes them all, the least-vetoed options are kept.
4. **Decide.** The Schulze winner (pairwise majority) among the remaining options is elected. Veto counts and the pairwise matrix are published.
5. **Revise, at most once.** Only if step 3 had to fall back: reopen nominations briefly, vote again; the second result is final.

What it guarantees (proved in the note): it **always decides**, in one round (two at most); the status quo gets **no default advantage**; the winner is **acceptable to more than two thirds** of the members whenever some option is; among acceptable options, **the majority decides**; and **adding variants of an option changes nothing**.

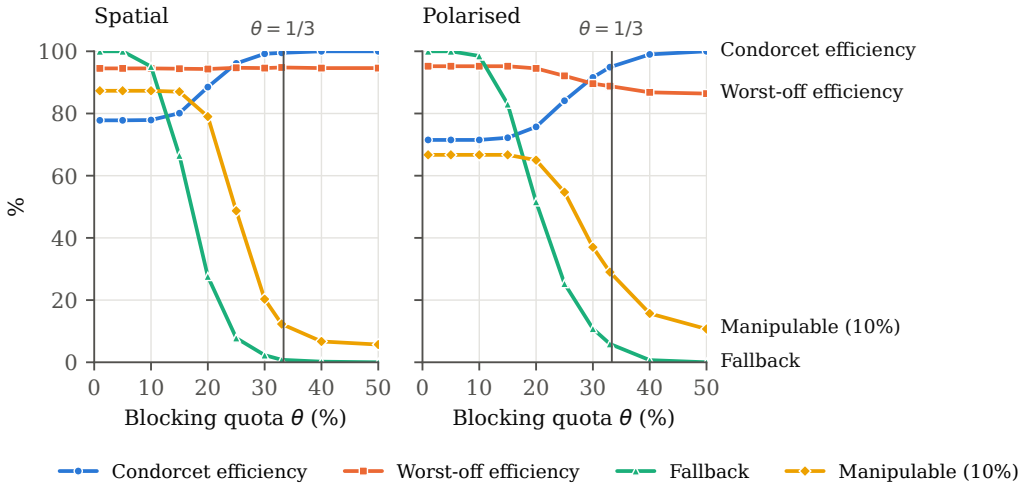
The evidence

Monte Carlo study of 21 rules, 2,000 simulated decisions per setting, 101 members, 8 options. Results for a polarised group (two blocs of 55% and 45%), the case in which a compromise matters most:

	Majority only (Schulze)	**QVC,** $\theta = 1/3$ (recommended)	**QVC,** $\theta = 1/4$ (more consensual)
Outcome rejected by at least a third of members	10.1%	5.9%	5.9%
Satisfaction of the least satisfied members (0–100)	85.2	88.4	92.8
Total satisfaction (0–100)	97.4	97.7	96.8
Majority's preferred option elected	100%	94.3%	83.2%
Manipulable by 10% of members	7%	27%	57%

Satisfaction: 0 for an average option, 100 for the best; the least satisfied member is the one at the 10th percentile. "Rejected by a third": 5.9% is the minimum any rule can reach, the share of decisions in which every option is rejected by a third. Unanimity with the status quo as default, in the same setting: deadlock in 97.9% of decisions.

Reading. With a blocking third, **contested outcomes are almost halved** and the least satisfied members do better, **at no cost in total satisfaction**; the majority's choice is set aside only when a third of the members reject it. Lowering the quota to a quarter protects minorities more but doubles the exposure to strategic blocking.



What the quota θ buys and costs. Below 20%, nearly every option is vetoed and the procedure falls back on the least-vetoed option; above 40%, vetoes almost never bind. One third balances the two.

The price, and how to contain it

Any blocking minority can also block an option it finds acceptable, to push its own favourite: no counting rule can prevent it (Greenberg, 1979). In our simulations a coalition of 10% of the members can manipulate 11–27% of decisions under QVC(1/3), against 5–7% under majority rule alone. Safeguards: **public vetoes with stated reasons** (a chair may void an unreasoned veto); **a quota no lower than one quarter**; **one revision round at most**; in bodies that decide often, **a budget of vetoes across decisions**, never within one.

Decisions for the organisation

- **Quota:** one third by default (with 10 members, 4 vetoes exclude an option); one quarter for a more consensual culture.
- **Status quo:** an ordinary option by default; for irreversible decisions, require the winner to beat it by a stated ratio (e.g. 2:1), as Debian does.
- **Revision round:** on or off, and its length.

Limits. The simulations use stylised electorates; the references were checked against publisher pages, not full texts. Next step: test the procedure on real ranked ballots (e.g. Debian's published votes).